02 TECHNIQUES IN AVOIDING AMBIGUITY

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TECHNIQUES AVOIDING AMBIGUITY

A collection of technical terms which are part of a language (terminology)

GLOSSARY

TECHNIQUES AVOIDING AMBIGUITY

TEMPLATE-BASED WORK PRODUCT

Using predefined structures for the requirements

innovative ● entrepreneurial ● global

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Type of Templates

TECHNIQUES AVOIDING AMBIGUITY: TEMPLATE-BASED



Phrase Templates

- Provide a predefined syntactic structure for a phrase that expresses a requirements.
- Use for writing an individual requirements OR user story



Form Templates

- Provide a set of predefined fields in a form to be filled
- Form template for
  a. Writing a use case
  b. A measurable quality requirements



Document Templates

- Provide a predefined structure for a requirements document

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02 (a – ii) TECHNIQUES IN AVOIDING AMBIGUITY: TEMPLATE-BASED

- *FORM TEMPLATES FOR USE CASES*

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Form Template for Writing Use Cases

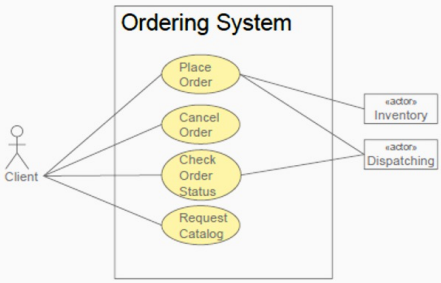
FORM TEMPLATE: USE CASES

- Use cases of a system and actors are visualised in a UML Use Case Diagram
- Form template for writing use case is an addition to the Use Case Diagram.
- Form template for writing use case is called a Use Case Specification

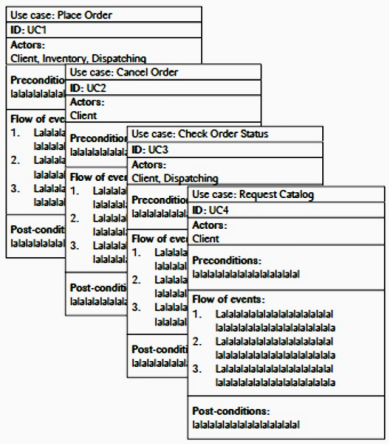
Use Case Specification

FORM TEMPLATE: USE CASES

Use Case Diagram



Use Case Specifications



Use Case Specification Template

FORM TEMPLATE: USE CASES

Use Case ID	<unique ID of the use case>
Use Case Name	<A short active verb phrase>
Description	<Short description of the use case>
Actor (s)	<Name of actor(s) involved>
Triggers	<Event that initiates the execution of the use case>
Pre-Condition(s)	<Condition(s) that must be true when the execution of the use case is triggered>
Normal Flow	<Description of the main success scenario in steps>
Alternative Flow(s)	<Description of alternative or option steps, with reference to the corresponding steps in the normal flow>
Exception Flow (s)	<Description of error handling steps, with reference to the corresponding steps in the normal or alternative flow>
Post Condition (s)	<Condition(s) that must be fulfilled if the use case ran successfully>
Related Requirements	<Use case(s) that this use case is connected to>

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Use Case Specification: Scenarios

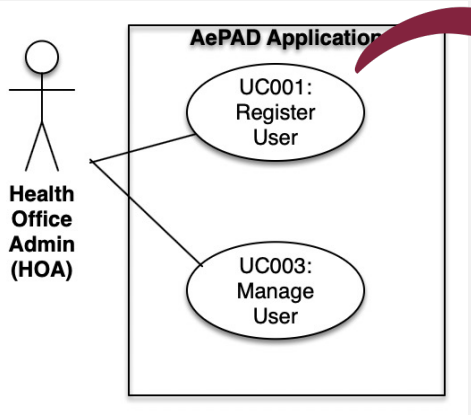
FORM TEMPLATE: USE CASES

- **E**very use case has a primary scenario
  - This is the “happy scenario”
  - In the use case specification, this scenario is written in the normal flow
- Every use case has multiple alternative scenarios
  - Alternative ways to success or options given for the scenarios. In the use case specification, this scenario is written in the Alternative Flow.
  - Exceptional scenarios to handle errors. In the use case specification, this scenario is written in the Exception Flow.
  - Both scenarios must have a reference to the corresponding steps in normal flow or alternative flow (for exception flow)

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Use Case Specification: Example 1

FORM TEMPLATE: USE CASES

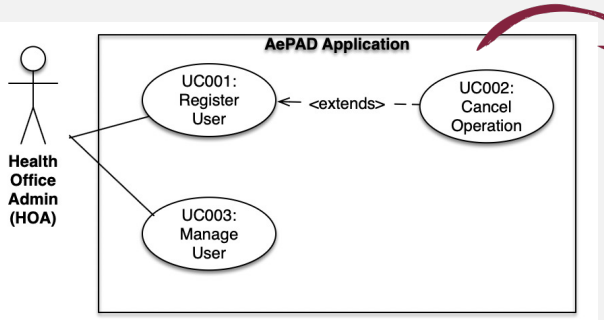


Use Case ID	UC001
Use Case Name	Register User
Actor (s)	Health Office Admin (HOA)
Description	This use case is used by HOA to register new user into the AePAD application
Triggers	HOA select register user
Pre-Condition(s)	HOA logged into the system
Normal Flow (NF)	1. The system displays register form 2. HOA enter new user information 3. HOA chooses "Confirm" and create new user 4. The system passes the form validation. (EF1) 5. The system performs database insert and update 6. The system success insert new user into database. (EF2) 7. Use case ends
Alternative Flow(s) (AF)	-
Exception Flow (s) (EF)	**1. Validation fail** 1.1. The system displays error message on the form 1.2. NF1 is executed again. **2. System failed to insert new record into database** 2.1. The system displays error message on the screen 2.2. NF1 is executed again.
Post Condition	HOA sucessfully register new user into the system
Related Requirements	-

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Use Case Specification: Example 2

FORM TEMPLATE: USE CASES

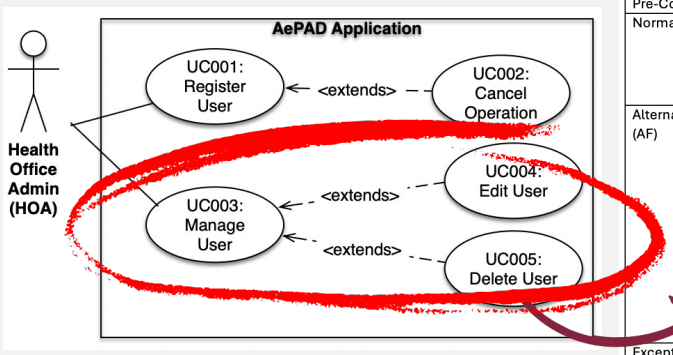


Use Case ID	UC001
Use Case Name	Register User
Actor (s)	Health Office Admin (HOA)
Description	This use case is used by HOA to register new user into the AePAD application
Triggers	HOA select register user
Pre-Condition(s)	HOA logged into the system
Normal Flow (NF)	1. The system displays register form 2. HOA enter new user information 3. HOA chooses "Confirm" and create new user 4. The system passes the form validation. (AF1, EF1) 5. The system performs database insert and update 6. The system success insert new user into database. (EF2) 7. Use case ends
Alternative Flow(s) (AF)	1. Cancel Operation 1.1. The system return back to the last accessed page. 1.2. The use case is aborted.
Exception Flow (s) (EF)	**1. Validation fail** 1.1. The system displays error message on the form 1.2. NF1 is executed again. **2. System failed to insert new record into database** 2.1. The system displays error message on the screen 2.2. NF1 is executed again.
Post Condition	HOA sucessfully register new user into the system
Related Requirements	-

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Use Case Specification: Example 3

FORM TEMPLATE: USE CASES

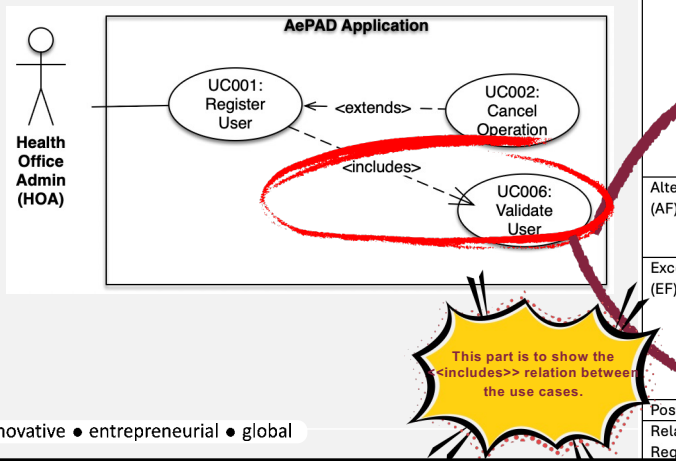


Use Case ID	UC003
Use Case Name	Manage User
Actor (s)	Health Office Admin (HOA)
Description	This use case is used by HOA to edit and delete the registered user in the AePAD application
Triggers	HOA select manage user
Pre-Condition(s)	HOA logged into the system
Normal Flow (NF)	1. The system displays user list 2. HOA enter choose the action to be performed. (AF1, AF2) 3. Use case ends
Alternative Flow(s) (AF)	1. Edit user 1.1. HOA choose the user to be edited. 1.2. The system display the chosen user form 1.3. HOA updates the user information 1.4. HOA choose "Confirm" and update user. 1.5. The system sucessfully update user information. (EF1) 2. Delete user 2.1. HOA choose the user to be deleted. 2.2. HOA choose to delete the user 2.3. The system succesfully delete the user and system update the user. (EF2)
Exception Flow (s) (EF)	**1. Update Fail** 1.1. The system displays error message on the screen 1.2. NF1 is executed again. **2. Delete Fail** 2.1. The system displays error message on the screen 2.2. NF1 is executed again.
Post Condition	HOA sucessfully update the user records in the system
Related Requirements	-

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Use Case Specification: Example 4

FORM TEMPLATE: USE CASES

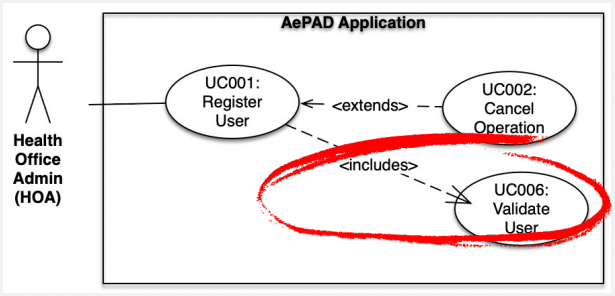


Use Case ID	UC001
Use Case Name	Register User
Actor (s)	Health Office Admin (HOA)
Description	This use case is used by HOA to register new user into the AePAD application
Triggers	HOA select register user
Pre-Condition(s)	HOA logged into the system
Normal Flow (NF)	1. The system displays register form 2. HOA enter new user information 3. HOA chooses "Confirm" and create new user 4. The system passes the form validation. Use case UC-006 Validate User is performed. (AF1, EF1) 5. The system performs database insert and update 6. The system success insert new user into database. (EF2) 7. Use case ends
Alternative Flow(s) (AF)	1. Cancel Operation 1.1. The system return back to the last accessed page. 1.2. The use case is aborted.
Exception Flow (s) (EF)	**1. Validation fail** 1.1. The system displays error message on the form 1.2. NF1 is executed again. **2. System failed to insert new record into database** 2.1. The system displays error message on the screen 2.2. NF1 is executed again.
Post Condition	HOA successfully register new user into the system
Related Requirements	UC006: Validate User

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Use Case Specification: Example 4 (cont.)

FORM TEMPLATE: USE CASES



Use Case ID	UC006
Use Case Name	Validate User
Actor (s)	Health Office Admin (HOA)
Description	This use case is used by the system to validate the user of AePAD application
Triggers	HOA click create new user
Pre-Condition(s)	HOA choose register new user
Normal Flow (NF)	1. The system receive the new user information 2. The system checks the information against the database. (EF1) 3. Use case ends
Alternative Flow(s) (AF)	-
Exception Flow (s) (EF)	**1. The user exists** 1.1. The system displays error message on the form 1.2. The use case abort.
Post Condition	The use case flow continues according to the flow of the UC001
Related Requirements	UC001: Register User

This part is to show the <<includes>> relation between the use cases.

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Misuse Case Specification Template

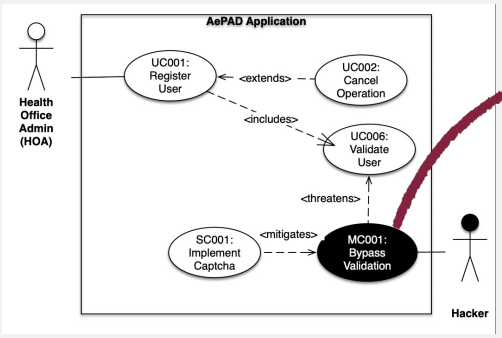
FORM TEMPLATE: USE CASES

Misuse Case ID	<unique ID of the misuse case>
Misuse Case Name	<A short active verb phrase>
Description	<Short description of the security threat or negative event>
Misuser/Threat Agent (s)	<Name of the unauthorized actor or external agent involved>
Threat	<The security vulnerability or flaw this misuse case exploits>
Triggers	<Event that initiates the execution of the misuse case (the attack start)>
Pre-Condition(s)	<Condition(s) that must be true before the misuse case can be executed (e.g., *System is online*, *Threat agent has network access*)>
Misuse Flow (Attack Steps)	<Description of the steps the misuser takes to execute the attack/harm>
Mitigation Flow(s)	<Description of alternative or option steps that could be taken by the system (or a legitimate user) to stop the attack mid-flow>
Exception Flow (s)	<Description of how the system handles critical errors during the attack and what prevents the attack from succeeding>
Post Condition (s)	<The resulting state of the system if the misuse case runs successfully (the harm done)>
Related Use Case(s)	<The legitimate use case(s) this misuse case threatens or is the inverse of>
Security Requirements	<The specific security requirement(s) that must be implemented to prevent this misuse case from occurring>

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Misuse Case Specification: Example

FORM TEMPLATE: USE CASES

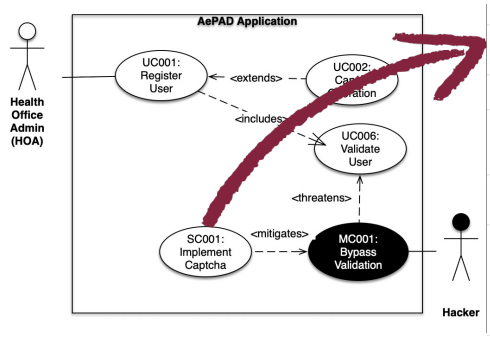


Misuse Case ID	MC001
Misuse Case Name	Bypass Validation
Description	1. An unauthorized actor defeats or circumvents the intended security check within the UC006: Validate User process. 2. The goal is to gain illegitimate access or register an unauthorized entity.
Misuser/Threat Agent	Hacker
Threat	The exploitation of a vulnerability (e.g., API manipulation, business logic flaw, or lack of proper server-side controls) to skip a required security validation step.
Triggers	1. The Hacker attempts to automate the execution of a related use case (like UC001: Register User) using a bot. 2. The Hacker manually intercepts and modifies application traffic during the validation sequence.
Pre-Condition(s)	1. The AePAD Application is running. 2. The Hacker has network access to the registration/login interface. 3. The target use case, UC006: Validate User, contains a detectable or exploitable flaw.
Misuse Flow (Attack Steps)	1. Hacker begins a legitimate use case (e.g., UC001) which includes UC006. 2. System triggers UC006: Validate User (e.g., sends a validation email or displays a challenge). 3. The Hacker performs an action to bypass the validation (e.g., sends a forged request, replays a valid session token, or exploits a sequence flaw). 4. The System, misled by the bypass, processes the request as if validation was successful.
Mitigation Flow(s)	1. SC001: Implement Captcha is triggered, forcing the Hacker to solve a challenge before initiating the flow, thereby mitigating automated attacks. 2. System detects the malicious attempt and blocks the session mid-flow.
Exception Flow(s)	1. If the System detects unusual activity (e.g., multiple failed attempts, rapid requests from one IP), it blocks the session. 2. The event is logged, and the targeted resource (e.g., the registration form) may be temporarily locked.
Post Condition(s)	1. The Hacker successfully completes the parent use case (UC001: Register User). 2. Unauthorized access to the application is gained without completing the necessary security validation.
Related Use Case(s)	1. UC006: Validate User (Threatened). 2. UC001: Register User (Leveraged).
Security Requirements	1. SR.Validation.1: All user input and validation status must be verified server-side. 2. SR.Auth.2: Critical endpoints must enforce strict authorization checks, independent of client-side validation state. 3. SR.Logging.3: Log and alert on all failed validation attempts exceeding a defined threshold.

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Security use Case Specification – use the same Use Case Specification Template

FORM TEMPLATE: USE CASES



Use Case ID	SC001
Use Case Name	Implement Captcha
Description	A system process to present a challenge (e.g., image recognition, mathematical problem) to a user/actor before sensitive operations to distinguish between human users and automated bots.
Actor (s)	Health Office Admin (HOA) (Indirectly, as a legitimate user) Hacker (The entity being challenged)
Triggers	1. The initiation of UC001: Register User that is susceptible to automated attack 2. The system detects suspicious activity (e.g., rapid attempts) that warrants an escalated security check.
Pre-Condition(s)	1. The security setting to enable CAPTCHA on the target use case is active. 2. The client is able to render and interact with the CAPTCHA interface.
Normal Flow	1. System displays the CAPTCHA challenge (e.g., "Select all squares with a traffic light"). 2. The User/Adversary attempts to solve the challenge. (A1) (E2) 3. The System verifies the solution. (E1) 4. If correct, the System allows the flow to proceed to the next step (UC006: Validate User).
Alternative Flow(s)	A1: The Hacker requests a new challenge 1. The hacker requests a new challenge (e.g., image refresh or audio challenge). 2. The System generates and presents a new challenge, returning to step 2 in Normal Flow.
Exception Flow (s)	E1: The solution is incorrect 1. The System displays an error message and increments a failure counter. 2. The System returns to step 1 in Normal Flow. E2: Failure Threshold: the failure counter exceeds a defined threshold (e.g., 3 incorrect attempts) 1. The System locks the session/IP address, preventing the use case from proceeding. 2. The use ends
Post Condition(s)	1. The malicious use case MC001: Bypass Validation is mitigated by preventing an automated attack. 2. A legitimate user is allowed to proceed to the next step of the use case (UC006: Validate User).
Related Requirements	1. SR.Validation.1: Must mitigate automated script registration attempts. 2. MC001: Bypass Validation (Directly mitigates this misuse case).
Related Use Case(s)	UC001: Register User UC006: Validate User

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